AUTOMOBILE ENGINEERING · PART 1 OF 11

The Engine, Part 1: How It Works

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

An engine is an air pump that burns fuel to push pistons. Each push turns a crankshaft, and that rotation eventually reaches the wheels. **CC** (cubic centimetres) is simply the total volume the pistons sweep: a measure of how much air the engine can breathe per cycle, not of how much power it makes. Power depends on how much air gets in, how much fuel is burnt with it, and how fast the engine spins. That is why a 1.2-litre turbo can out-pull a 1.5-litre non-turbo.

1. Where the engine sits in the car

In a typical Indian petrol or diesel car the engine is a single block of metal under the bonnet, bolted to a gearbox. Everything else in the car (suspension, brakes, electronics) exists to carry, steer and stop what the engine moves. For a manager it is the single costliest mechanical part and the one most shaped by regulation (BS6 emission norms) and tax (engine size and fuel type affect the slab a car falls in).

2. The parts that matter

Five parts explain nearly everything:

- **Cylinder:** a sealed chamber where fuel burns. Cars have 3 or 4 in India; a few have more.
- **Piston:** a plug inside the cylinder, pushed down by the burning gas.
- **Connecting rod and crankshaft:** convert the piston's up-down motion into rotation.
- **Valves:** doors at the top of the cylinder. Intake valves let air (and fuel) in; exhaust valves let burnt gas out.
- **Spark plug (petrol) or fuel injector (diesel):** starts the combustion.

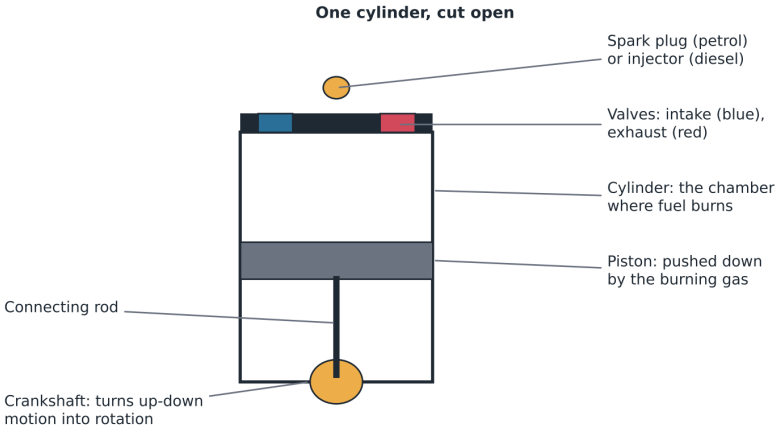


Fig 1. One cylinder, cut open. Everything else in the engine is repetition and refinement of this.

3. The four-stroke cycle

Almost every car engine repeats the same four-step loop thousands of times a minute. A *stroke* is one trip of the piston up or down, so one full cycle takes four strokes and two turns of the crankshaft.

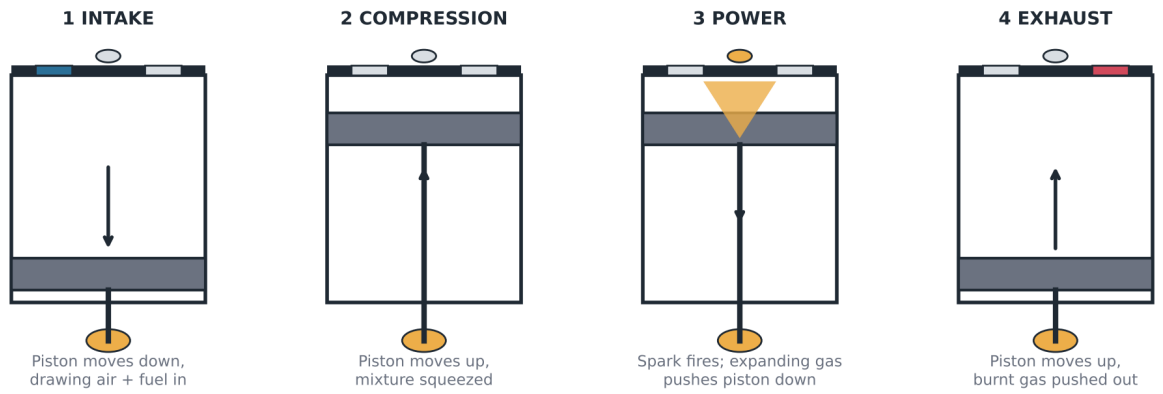


Fig 2. Intake, compression, power, exhaust. Only the third stroke produces work; the other three are paid for by the crankshaft's momentum (and by other cylinders).

- **Intake:** piston moves down; the intake valve opens and air (plus fuel, in a petrol engine) is drawn in.
- **Compression:** both valves shut; the piston moves up and squeezes the mixture. More squeezing means more energy per burn, which is why compression ratio matters.
- **Power:** a spark (petrol) or the heat of compressed air igniting injected fuel (diesel) triggers combustion; the expanding gas drives the piston down.
- **Exhaust:** the exhaust valve opens and the rising piston pushes burnt gas out.

4. What CC actually means

Each cylinder's **swept volume** is the area of the piston face times the distance it travels (the stroke): **π/4 × bore² × stroke**. Multiply by the number of cylinders and you get engine capacity.

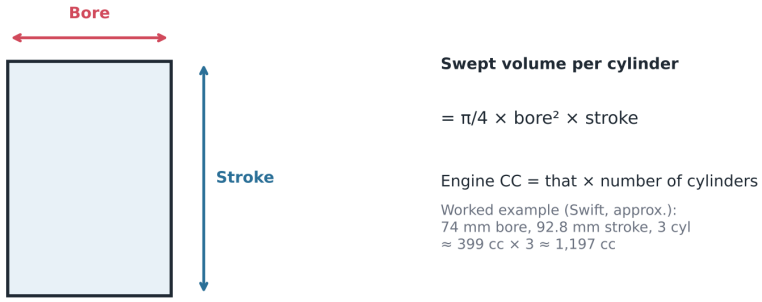


Fig 3. Bore is the cylinder's width; stroke is the piston's travel. Swift figures are third-party database values and should be treated as approximate.

Take the Maruti Suzuki Swift's Z12E engine: 1,197 cc across 3 cylinders, so each cylinder sweeps about 399 cc. It makes 80.46 bhp at 5,700 rpm and 111.7 Nm at 4,300 rpm.

Same CC, very different engines. The Tata Nexon's 1.2 Revotron turbo is 1,199 cc, effectively identical to the Swift's. Yet it makes 118.27 bhp and 170 Nm, about 47% more power and 52% more torque from the same cylinder volume. The reason is the turbocharger: it packs more air into each cylinder, so each cycle burns more fuel. CC measures the size of the lungs, not the athlete.

CC and tax. India has historically taxed cars partly by engine size and length (the sub-1,200 cc petrol / sub-1,500 cc diesel, sub-4 m category was the benefit slab). Rates changed with the September 2025 GST revision, so check the current slab before using engine size as a tax shortcut.*

5. Cylinders: why 3, 4 or 1

For a given CC, more cylinders means smaller, lighter pistons that fire more often, which gives a smoother, quieter engine that revs freely. Fewer cylinders means fewer parts, less friction and lower cost, but more vibration. Indian small cars mostly use three cylinders for cost and efficiency; four-cylinder engines appear where refinement is a selling point (the Creta's 1.5 MPi is a 4-cylinder, 1,497 cc, 113 bhp unit).

The Swift's move from the older K12 four-cylinder to the three-cylinder Z12E shows the trade-off: lighter, cheaper and more efficient, in return for a slightly less smooth idle. **Valves** add a further lever: most modern engines use four valves per cylinder (two intake, two exhaust) because more door area lets more air in and out.

6. Petrol versus diesel

Both use the four strokes; they differ in how combustion starts. A petrol engine mixes fuel with air and ignites it with a spark. A diesel engine compresses air alone, far harder (compression ratio roughly 16–18:1 versus roughly 10–13:1 for petrol), until it is hot enough to ignite fuel sprayed in.

Diesel's high compression gives more torque at low rpm and better fuel economy, but the engine is heavier, and BS6 emission control (particulate filter, urea-based SCR) adds cost. The Creta shows it clearly: the 1.5 diesel (1,493 cc) makes 114 bhp but 250 Nm from 1,500 rpm; the 1.5 petrol MPi (1,497 cc) has almost the same cc and 113 bhp, but only 143.8 Nm. Same size, similar power, 74% more torque.

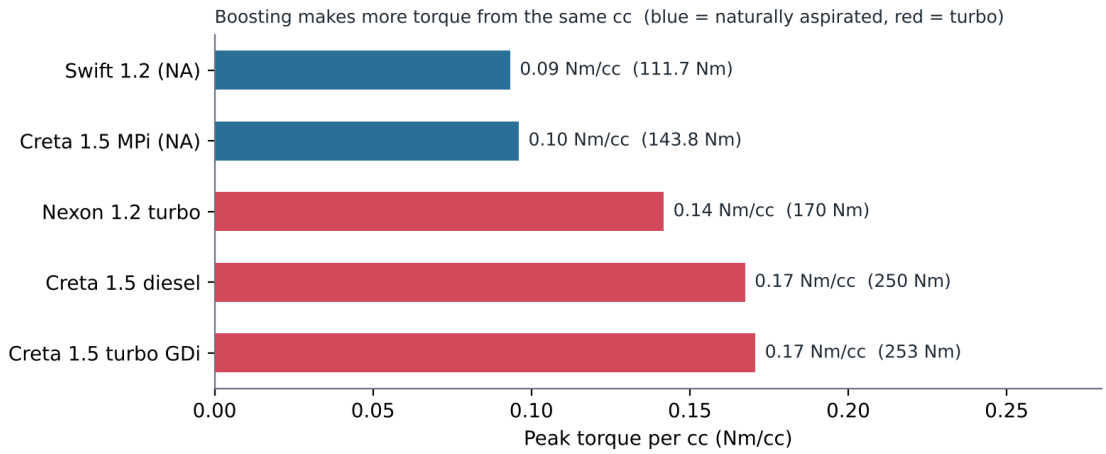


Fig 4. Torque per cc for five engines in this article. Boosted engines (red) deliver roughly 0.14–0.17 Nm per cc against about 0.09–0.10 for the plain petrols (blue).

7. The Indian engine line-up in one table

Car / engine	CC	Cyl	Power (bhp)	Torque (Nm)	Aspiration
Swift 1.2 Z12E	1,197	3	80.46 @ 5,700	111.7 @ 4,300	Natural
Nexon 1.2 Revotron	1,199	3	118.27 @ 5,500	170 @ 1,750–4,000	Turbo
Creta 1.5 MPi	1,497	4	113 @ 6,300	143.8 @ 4,500	Natural
Creta 1.5 diesel	1,493	4	114 @ 4,000	250 @ 1,500–2,750	Turbo diesel
Creta 1.5 turbo GDi	1,482	4	158 @ 5,500	253 @ 1,500–3,500	Turbo
Thar 2.0 mStallion	1,997	4	~150	300–320	Turbo
Thar 2.2 mHawk	2,184	4	~130	~300	Turbo diesel

Five of these seven engines are turbocharged (Nexon, Creta diesel, Creta turbo, and both Tharas). They make about 0.14–0.17 Nm per cc against roughly 0.09–0.10 for the two naturally aspirated

petrols (Swift, Creta MPi). The modern trend is simple: shrink the engine, add a turbo, and keep the power.

8. Japanese and German engines in India

Japanese and German makers solve the same physics with different philosophies, and both are well represented here. Japanese brands (Maruti Suzuki, Toyota, Honda) dominate the mass market. German-group brands sit in the mid-market (Skoda and Volkswagen) and at the top (BMW, Mercedes-Benz, Audi).

Car / engine	Origin	CC	Cyl	Power	Torque (Nm)	Approach
Honda City 1.5 i-VTEC	Japan	1,498	4	~119 bhp (121 PS)*	145*	Natural, high-revving
Toyota Innova Hycross 2.0 petrol	Japan	1,987	4	~173 bhp	204	Natural
Toyota Innova Hycross 2.0 hybrid	Japan	1,987	4	~184 bhp system	~188 (engine)	Natural + electric motor
Skoda Kushaq / Slavia 1.0 TSI	Germany (VW Group)	999	3	115 PS	178*	Small turbo
Skoda Kushaq / Slavia 1.5 TSI	Germany (VW Group)	1,498	4	150 PS	250*	Turbo, cylinder deactivation*
BMW 330Li	Germany	1,998	4	258 hp	400	Turbo
Mercedes-Benz C 220d	Germany	1,993	4*	197 bhp	440	Turbo diesel + 48V mild hybrid

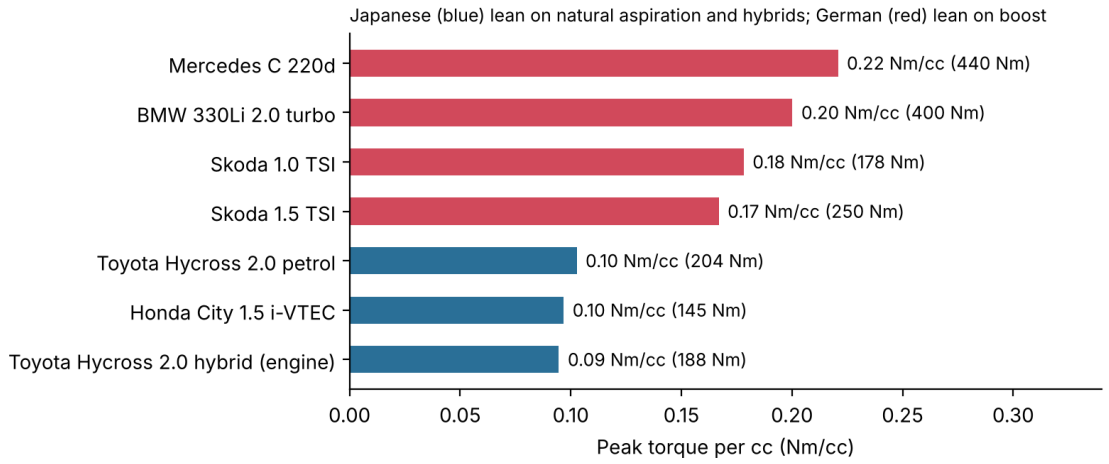


Fig 5. Torque per cc. Japanese naturally aspirated engines sit near 0.10 Nm per cc; German boosted engines run 0.17–0.22.

The Japanese way: breathe naturally, then add electric help. Honda's 1.5 i-VTEC makes about 0.10 Nm per cc with no turbo. It needs revs to deliver its power, but it is simple, smooth and famously durable. Toyota's 2.0-litre goes further: the hybrid version adds an electric motor to the same engine, chasing efficiency rather than peak torque. Maruti's Swift (Section 4) follows the same logic: small, light, cheap to run.

The German way: shrink the engine, add a turbo. Skoda's 1.0 TSI is a three-cylinder of just 999 cc, yet its torque per cc (0.18) is nearly double the Honda's. The 1.5 TSI can shut down two of its four cylinders at light load to save fuel.* BMW's 2.0-litre turbo petrol and Mercedes' 2.0 diesel reach 0.20 and 0.22 Nm per cc. Mercedes also adds a 48-volt mild-hybrid system that gives a short boost of up to 20 hp and 200 Nm.

Neither is simply better. Boosted engines pull hard at low rpm from small capacity, which suits city and highway driving, but run hotter and have more parts to maintain. Naturally aspirated engines are simpler and reward revs, but need more cc for the same pull. Part 2 explains how to choose between them for the way you drive.

Summary

- An engine burns fuel in cylinders; pistons turn a crankshaft; the crankshaft drives the wheels.
- Four strokes: intake, compression, power, exhaust. Only one produces work.
- CC = π/4 × bore² × stroke × cylinders. It measures size, not power.
- Same CC can mean very different outputs: Swift vs Nexon (1.2 L, 80 vs 118 bhp).
- 3 cylinders: cheaper and efficient; 4 cylinders: smoother.
- Diesel compresses air harder, giving more low-rpm torque but costlier BS6 hardware.
- Five of seven engines in our sample are boosted, which is the market's direction.
- Japanese makers lean on natural aspiration and hybrids (about 0.10 Nm/cc); German makers lean on small turbos (0.17–0.22 Nm/cc).

*Unverified or approximate: GST thresholds (18% versus 40% from 22 September 2025); Swift Z12E bore, stroke and compression ratio from third-party databases; compression-ratio ranges and efficiency rules of thumb are textbook generalisations. Skoda 1.0/1.5 TSI torque and cylinder deactivation, Honda City figures and the Mercedes C 220d cylinder count are from listings or general knowledge, not an official brochure. Specs: manufacturer brochures, press kits and listings, November 2026. Theory: Heywood, Internal Combustion Engine Fundamentals.

See also: Automobile Engineering 2/11 — The Engine, Part 2: power, torque and mileage.